

Pfutsaro Phek Road - Summary of Cost

Bill No.	Description	Amount (Rs.)	Figure in Crore
A	Civil Works		
1	SITE CLEARANCE AND DISMANTALING	1,53,18,649	1.53
2	EARTH WORKS	1,23,12,45,505	123.12
3	GRANULAR SUB-BASE AND BASE COURSES	1,31,77,33,700	131.77
4	BITUMINOUS COURSES AND CONCRETE PAVEMENTS	68,63,33,109	68.63
5	CROSS DRAINAGE WORKS	1,70,75,62,820	170.76
6	BRIDGES / UNDERPASS	6,16,38,010	6.16
7	REPAIR OF EXISTING STRUCTURES	0	0.00
8	DRAINAGE AND PROTECTIVE WORKS	3,21,18,65,097	321.19
9	TRAFFIC SIGNS, MARKINGS AND ROAD APPURTENANCES	17,47,90,136	17.48
10	MISCELLANEOUS	6,78,61,785	6.79
Sub Total for Civil Works (A)		8,47,43,48,810	847.43
B	Utility Shifting Cost	1,30,00,000	1.30
Sub Total (A+B)		8,48,73,48,810	848.73
C	Physical Contingencies @5% on (A+B)	42,43,67,441	42.44
Grand Total(A+B+C)		8,91,17,16,251	891.17
Length of the package Road (km)		62.6	Km
Cost per km on TPC		14.2	Crore

Preparation of Revised DPR and Verification of Executed Quantities/Items for widening to 2-lane of 4 roads in state of Nagaland under SARDP-NE

RTES Ltd.
Highway Division

**BILL OF QUANTITIES (BOQ)
BILL NO. 1:GENERAL AND SITE CLEARANCE**

Item No	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
1.01	Clearing and Grubbing of Roadway as per Technical Specification Clause 201.	Ha	175	55,705	97,53,461
1.02	Dismantling of Existing Structures or any Existing Civil Works within Right of Way, as per Technical Specification Clause 202.				
	(i) Brick Masonry	cum	Nil	357	-
	(ii) Stone Masonry	cum	8,080	357	28,84,560
	(iii) Lime/Plain Cement Concrete in all grades	cum	700	542	3,79,400
	(iv) Reinforced / Pre-stressed Cement Concrete in all grades	cum	120	916	1,09,920
	(v) Bituminous courses	cum	1,966	336	6,60,724
	(vi) Non Bituminous courses	cum	Nil	578	-
	(vii) Cement concrete Pavements	cum	Nil	1,776	-
	(viii) Ordinary KM stone/Guard stone/Sign Post	No.	5	253	1,265
	(ix) Hectometre Stone	No.	Nil	51	-
	(x) 5th KM stone	No.	Nil	435	-
	(xi) Kerb stone	m	Nil	17	-
	(xii) All types of Fencing including posts, foundation concrete, back filling of pit etc. complete.	m	Nil	55	-
	(xiii) RCC Railing	m	Nil	258	-
	(xiv) Guard Rails/W Beam Crash Barrier	m	Nil	87	-
	(xv) Brick Pitching / Brick soling	cum	Nil	229	-
	(xvi) Removal of abandoned Telephone / Electric Poles and Lines including excavation and dismantling of foundation concrete and lines under the supervision of concerned department.	m	Nil	194	-
	(xvii) Removing all types of hume pipes and stacking serviceable material with all leads & lifts including earthwork and dismantling of masonry works.				
	(a) Up to 600 mm dia		Nil		-
	(b) Above 600 mm to 900 mm dia	m	Nil	206	-
	(c) Above 900 mm dia	m	2,220	279	6,19,380
	(xviii) Dismantling stone pitching/ dry stone spalls	m	Nil	478	-
	(xix) Dismantling of CI Water Pipe Line up to 1000 mm dia.	cum	Nil	245	-
		m	Nil	146	-
Carried to Page Collection					1,44,08,710

Annexure 2.1 (d) : Template Shape / Component Report (from Mx Roads) : PP Road											Cut Slope			
Station	Cut Volume	Fill Volume	BC Volume	Tack Coat Sur. Area	DBM Volume	Prime Coat Sur. Area	WMM Volume	GSB Volume	Subgrade Volume	<6m Sur. Area	<12 m Sur. Area	> 12 m Sur. Area	R Wall Height	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	
0														
10	129.1	5.9	3.1	103.3	5.2	104.0	26.8	23.4	54.1	11.0			1.7	
20	115.1	12.5	3.0	100.0	5.0	100.0	25.5	21.9	54.1	15.0			1.9	
30	126.8	13.2	3.0	100.0	5.0	100.0	25.5	21.9	54.1	29.0			1.7	
36	81.7	6.6	1.7	56.7	2.9	58.0	14.5	12.5	30.7	24.0			1.6	
40	65.7	4.3	1.3	43.3	2.2	44.0	11.0	9.5	23.3	21.0			1.5	
50	175.0	8.2	3.1	103.3	5.2	104.0	26.2	22.5	55.3	58.0			1.3	
51	13.5	0.5	0.2	6.7	0.4	8.0	1.8	1.6	3.9	4.0			1.3	
55	99.1	3.3	1.4	46.7	2.4	48.0	12.3	10.5	25.8		35.0		1.3	
60	116.8	3.6	1.6	53.3	2.6	52.0	13.3	11.4	28.0		76.0		1.4	
70	202.7	10.9	3.3	110.0	5.5	110.0	27.7	23.8	58.3	31.0			1.8	
80	144.5	15.4	3.3	110.0	5.5	110.0	27.7	23.8	58.3	54.0			2.0	
88	66.2	13.9	2.6	86.7	4.3	86.0	21.6	18.5	45.5	25.0			2.0	
90	8.5	4.6	0.7	23.3	1.2	24.0	5.9	5.1	12.5	3.0			2.1	
92	7.8	5.9	0.7	23.3	1.2	24.0	6.2	5.3	13.1	2.0			2.2	
100	22.4	27.1	2.4	80.0	3.9	78.0	20.0	17.1	42.2	11.0			2.7	
107	21.6	26.2	2.2	73.3	3.7	74.0	18.7	16.1	39.6	11.0			2.2	
110	8.2	8.3	0.8	26.7	1.3	26.0	6.8	5.9	14.5	2.0			2.4	
120	25.2	48.1	3.0	100.0	5.0	100.0	25.5	21.9	54.0	4.0			3.1	
130	25.5	54.6	3.0	100.0	5.0	100.0	25.5	21.9	54.0	5.0			2.8	
140	59.9	30.7	3.0	100.0	5.0	100.0	25.5	21.9	53.9	22.0			1.7	
150	106.3	10.3	3.0	100.0	5.0	100.0	25.5	21.9	53.9	45.0			1.2	
160	185.7	6.2	3.1	103.3	5.1	102.0	26.0	22.3	54.7		74.0		1.0	
161	22.5	0.5	0.3	10.0	0.5	10.0	2.4	2.0	5.0		13.0		1.0	
161	2.6	0.1			0.1	2.0	0.3	0.2	0.6		2.0		1.3	
161													1.3	
170	222.6	10.1	2.9	96.7	4.8	96.0	24.5	21.0	51.7		137.0		1.7	
176	101.5	9.3	2.0	66.7	3.3	66.0	16.6	14.3	35.0	14.0			2.0	
180	39.5	8.2	1.3	43.3	2.2	44.0	11.1	9.5	23.4	17.0			2.2	
180	4.0	1.2	0.2	6.7	0.3	6.0	1.4	1.2	2.9	2.0			2.2	
190	91.9	17.0	3.1	103.3	5.2	104.0	26.4	22.6	55.4	38.0			1.7	
200	114.3	13.7	3.4	113.3	5.7	114.0	29.0	25.4	64.1	38.0				
210	101.6	8.1	3.5	116.7	5.8	116.0	30.4	27.0	69.9	31.0				
220	80.9	0.4	3.5	116.7	5.8	116.0	30.4	27.0	69.9	19.0				
230	84.2	0.4	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
240	96.3	0.7	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
250	99.0	0.5	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
260	96.1	0.2	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
270	91.2	0.8	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
280	87.6	1.2	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
290	69.5	4.6	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
300	50.1	14.2	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
302	8.9	3.8	0.6	20.0	1.1	22.0	5.6	5.0	12.8					
310	40.0	14.9	2.8	93.3	4.8	96.0	24.8	22.0	56.9					
320	51.9	14.0	3.5	116.7	5.8	116.0	30.3	26.9	69.3					
328	44.2	13.0	2.8	93.3	4.6	92.0	24.1	21.3	54.9					
330	11.2	4.3	0.7	23.3	1.2	24.0	6.2	5.5	14.1					
340	52.9	22.3	3.5	116.7	5.8	116.0	30.2	26.8	68.8					
350	57.7	16.6	3.5	116.7	5.8	116.0	30.2	26.8	68.8					
360	78.6	7.4	3.5	116.7	5.8	116.0	30.2	26.8	68.8	1.0				
370	107.8	5.0	3.4	113.3	5.7	114.0	29.0	25.3	58.6	4.0			1.0	
380	100.0	11.1	3.3	110.0	5.5	110.0	27.8	23.8	58.6	5.0			2.1	
390	77.7	30.9	3.3	110.0	5.5	110.0	27.8	23.8	58.6	6.0			3.5	
400	79.9	48.0	3.3	110.0	5.5	110.0	27.8	23.8	58.6	8.0			3.7	
410	80.6	53.5	3.3	110.0	5.5	110.0	28.0	24.0	59.0	8.0			3.8	
413	21.2	16.7	0.9	30.0	1.5	30.0	7.8	6.7	16.5	2.0			4.0	
413	0.7	0.6			0.1	2.0	0.3	0.2	0.6				3.9	
413													3.9	
420	43.8	46.3	2.4	80.0	4.0	80.0	20.2	17.3	42.5	3.0			4.2	
428	38.0	59.7	2.6	86.7	4.4	88.0	22.4	19.2	47.2	1.0			4.8	
430	8.8	18.2	0.7	23.3	1.2	24.0	6.1	5.2	12.7				5.0	
440	33.5	77.0	3.4	113.3	5.6	112.0	28.5	24.4	59.8				4.4	
440	1.0	2.4	0.1	3.3	0.2	4.0	1.1	0.9	2.2				4.3	
450	26.4	37.0	3.2	106.7	5.4	108.0	27.4	23.5	57.5				1.9	
460	40.3	9.8	3.4	113.3	5.6	112.0	28.5	24.4	59.7	2.0			1.0	
470	75.8	5.6	3.4	113.3	5.6	112.0	28.6	24.5	59.9	12.0			1.0	
479	95.7	4.9	3.0	100.0	5.0	100.0	25.1	21.5	52.6	17.0			1.0	
479	1.3	0.1			0.1	2.0	0.3	0.2	0.6				1.2	
479													1.5	
480	16.7	1.6	0.4	13.3	0.7	14.0	3.4	2.9	7.2	3.0			1.5	
490	130.8	15.0	3.4	113.3	5.7	114.0	29.1	24.9	61.2	26.0			1.6	
500	103.5	20.5	3.5	116.7	5.8	116.0	29.3	25.1	61.6	21.0			1.7	
510	72.8	28.6	3.5	116.7	5.8	116.0	29.3	25.1	61.6	16.0			1.9	
520	51.2	37.0	3.5	116.7	5.8	116.0	29.3	25.1	61.6	10.0			2.2	
530	36.5	45.6	3.5	116.7	5.8	116.0	29.3	25.1	61.6	6.0			2.5	
540	26.4	54.6	3.5	116.7	5.8	116.0	29.3	25.1	61.6	4.0			2.9	
550	18.8	64.4	3.5	116.7	5.8	116.0	29.3	25.1	61.6	2.0			3.2	
560	14.1	72.1	3.5	116.7	5.8	116.0	29.3	25.1	61.6	1.0			3.2	
570	11.4	76.8	3.5	116.7	5.8	116.0	29.3	25.1	61.6				3.3	
580	16.8	79.6	3.5	116.7	5.8	116.0	29.3	25.1	61.6	4.0			3.5	
590	38.6	75.4	3.5	116.7	5.8	116.0	29.3	25.1	61.6	14.0			3.5	
592	12.9	15.3	0.8	26.7	1.3	26.0	6.7	5.7	14.0	5.0			3.3	
600	60.7	37.0	2.7	90.0	4.5	90.0	22.6	19.4	47.5	19.0			2.4	
607	91.6	17.8	2.5	83.3	4.2	84.0	21.3	18.4	44.8	23.0			1.6	
610	44.5	4.3	0.9	30.0	1.6	32.0	8.0	6.8	16.8	10.0			1.6	
620	196.7	13.1	3.5	116.7	5.8	116.0	29.3	25.1	61.7	39.0			1.4	
621	17.5	0.9	0.3	10.0	0.5	10.0	2.3	2.0	4.8	3.0			1.4	
630	257.6	9.5	3.3	110.0	5.5	110.0	28.1	24.5	61.5	44.0				
640	389.2	4.8	3.6	120.0	6.0	120.0	30.7	27.0	63.6	58.0				
650	491.8	2.7	3.5	116.7	5.8	116.0	29.5	25.6	63.6		82.0		1.0	
653	186.7	2.0	1.2	40.0	2.0	40.0	10.1	8.7	21.3		58.0		1.0	
660	355.2	8.4	2.3	76.7	3.8	76.0	19.2	16.4	40.4		117.0		1.9	
670	509.3	19.9	3.5	116.7	5.8	116.0	29.3	25.1	61.6		144.0		1.7	
680	538.0	15.1	3.5	116.7	5.8	116.0	29.3	25.1	61.5		203.0		1.3	
685	331.4	4.5	1.9	63.3	3.1	62.0	15.7	13.4	32.9		166.0		1.2	
690	312.1	3.1	1.6	53.3	2.7	54.0	13.6	11.6	28.4		148.0		1.1	
700	706.3	6.2	3.5	116.7	5.8	116.0	29.2	25.0	61.3		331.0		1.1	
710	661.2	7.5	3.5	116.7	5.8	116.0	29.2	25.0	61.3		354.0		1.7	
720	613.9	16.0	3.5	116.7	5.8	116.0	29.2	25.0	61.3		290.0		2.4	
726	416.8	11.6	2.0	66.7	3.3	66.0	16.7	14.3	35.0			150.0	2.1	
730	375.7	6.1	1.5	50.0	2.5	50.0	12.6	10.8	26.3			229.0	1.7	
740	1083.2	7.8	3.5	116.7	5.8	116.0	29.2	25.0	61.3			418.0	0.9	
750	1587.7	3.5	3.5	116.7	5.8	116.0	29.2	25.0	61.2			506.0	0.5	
756	1061.8	2.1	2.0	66.7	3.3	66.0	16.7	14.3	34.9			412.0	1.1	
760	706.5	3.2	1.5	50.0	2.5	50.0								

QUANTITY CALCULATION FOR 1X1.2 DIA PIPES

S. No.	Proposed Chainage (km)	Designation	Pipe Length	Excavation	Back Filling	Conc. Foundation	Conc. Substructure	Levelling coarse	Filter Media	Painting	Lettering
1	0.925	1*1.2	12.500	102.174	137.214	12.96	20.4	3.472	30.24	2.28	50
2	8.050	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
3	30.940	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
4	46.415	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
5	47.050	1*1.2	12.500	165.490	179.180	14.58	24	3.875	33.84	2.28	50
6	57.430	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
7	43.950	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
8	49.670	1*1.2	12.500	137.415	137.214	12.96	20.4	3.472	30.24	2.28	50
9	15.150	1*1.2	12.500	137.415	137.214	12.96	20.4	3.472	30.24	2.28	50
10	33.510	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
11	48.965	1*1.2	12.500	165.490	179.180	14.58	24	3.875	33.84	2.28	50
12	60.595	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
13	61.820	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
14	54.240	1*1.2	12.500	137.779	137.214	12.96	20.4	3.472	30.24	2.28	50
15	57.770	1*1.2	10.000	137.144	137.214	12.96	20.4	3.472	30.24	2.28	50
16	57.840	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
17	30.760	1*1.2	12.500	133.214	137.214	12.96	20.4	3.472	30.24	2.28	50
18	12.910	1*1.2	12.500	163.543	179.180	14.58	24	3.875	33.84	2.28	50
19	22.800	1*1.2	12.500	143.779	137.214	12.96	20.4	3.472	30.24	2.28	50
20	50.510	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
21	12.210	1*1.2	12.500	129.215	137.214	12.96	20.4	3.472	30.24	2.28	50
22	50.475	1*1.2	12.500	127.812	137.214	12.96	20.4	3.472	30.24	2.28	50
23	45.640	1*1.2	12.500	127.604	137.214	12.96	20.4	3.472	30.24	2.28	50
24	52.970	1*1.2	10.000	131.343	137.214	12.96	20.4	3.472	30.24	2.28	50
25	59.630	1*1.2	12.500	125.924	137.214	12.96	20.4	3.472	30.24	2.28	50

S. No.	Proposed Chainage (km)	Designation	Pipe Length	Excavation	Back Filling	Conc. Foundation	Conc. Substructure	Levelling coarse	Filter Media	Painting	Lettering
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QUANTITY CALCULATION FOR 2X1.2 DIA PIPES

236	27.490	2*1.2	25.000	132.630	137.214	18.150	21.74279674	3.596	30.24	2.28	50
237	5.500	2*1.2	25.000	132.567	137.214	18.150	21.74279674	3.596	30.24	2.28	50
238	41.300	2*1.2	25.000	127.608	137.214	18.150	21.74279674	3.596	30.24	2.28	50
239	21.450	2*1.2	25.000	127.185	137.214	18.150	21.74279674	3.596	30.24	2.28	50
240	2.950	2*1.2	25.000	157.743	179.180	20.525	26.24279674	3.968	33.84	2.28	50
241	3.260	2*1.2	20.000	128.347	137.214	18.150	21.74279674	3.596	30.24	2.28	50
242	3.200	2*1.2	25.000	127.461	137.214	18.150	21.74279674	3.596	30.24	2.28	50
243	3.122	2*1.2	25.000	125.367	137.214	18.150	21.74279674	3.596	30.24	2.28	50
244	13.720	2*1.2	25.000	120.233	137.214	18.150	21.74279674	3.596	30.24	2.28	50
245	3.680	2*1.2	25.000	148.032	179.180	20.525	26.24279674	3.968	33.84	2.28	50
246	6.165	2*1.2	25.000	118.663	137.214	18.150	21.74279674	3.596	30.24	2.28	50
247	16.760	2*1.2	25.000	117.945	137.214	18.150	21.74279674	3.596	30.24	2.28	50
248	4.445	2*1.2	25.000	145.396	179.180	20.525	26.24279674	3.968	33.84	2.28	50
249	21.555	2*1.2	25.000	116.588	137.214	18.150	21.74279674	3.596	30.24	2.28	50
250	25.920	2*1.2	25.000	116.475	137.214	18.150	21.74279674	3.596	30.24	2.28	50
251	44.465	2*1.2	25.000	113.938	137.214	18.150	21.74279674	3.596	30.24	2.28	50
252	4.090	2*1.2	25.000	140.290	179.180	20.525	26.24279674	3.968	33.84	2.28	50
253	17.000	2*1.2	25.000	111.405	137.214	18.150	21.74279674	3.596	30.24	2.28	50
254	6.820	2*1.2	25.000	137.753	179.180	20.525	26.24279674	3.968	33.84	2.28	50

S. No.	Proposed Chainage	Designation	Excavation	RCC for box	Approach slab RCC	Exp. Gap	Lettering	Drainage Spout	Painting	Levelling Course	Excavation for Return Wall	RCC for Return Wall	Filter Media	Backfill	RCC for Counter fort	RCC for Wall
	(km)		m ³	m ³	m ³	lm	nos.	nos.	m ²	m ³	m ³	m ³	m ³	m ³	m ³	m ³
17	45.685	1.22.00	90.09	56.51	26.71	21.80	50.00	4.00	9.88	15.87	70.20	22.24	20.16	91.48	0.00	0.00
18	40.465	1.22.00	89.34	56.51	26.71	21.80	50.00	4.00	9.88	15.87	69.83	22.24	20.16	91.48	0.00	0.00
19	55.970	1.22.00	91.70	57.95	27.44	22.40	50.00	4.00	9.88	16.30	69.78	22.24	20.16	91.48	0.00	0.00
20	1.260	1.22.00	91.43	57.95	27.44	22.40	50.00	4.00	9.88	16.30	69.65	22.24	20.16	91.48	0.00	0.00
21	2.090	1.22.00	88.01	56.51	26.71	21.80	50.00	4.00	9.88	15.87	69.16	22.24	20.16	91.48	0.00	0.00
22	39.260	1.22.00	84.70	55.07	25.97	21.20	50.00	4.00	9.88	15.44	68.70	22.24	20.16	91.48	0.00	0.00
23	22.470	1.22.00	87.89	59.39	28.18	23.00	50.00	4.00	9.88	16.73	66.79	22.24	20.16	91.48	0.00	0.00
24	47.870	1.22.00	86.67	56.51	26.71	21.80	50.00	4.00	9.88	15.87	68.48	22.24	20.16	91.48	0.00	0.00
25	51.140	1.22.00	91.18	59.39	28.18	23.00	50.00	4.00	9.88	16.73	68.36	22.24	20.16	91.48	0.00	0.00
26	42.250	1.22.00	86.20	56.51	26.71	21.80	50.00	4.00	9.88	15.87	68.25	22.24	20.16	91.48	0.00	0.00
27	47.300	1.22.00	90.92	59.39	28.18	23.00	50.00	4.00	9.88	16.73	68.23	22.24	20.16	91.48	0.00	0.00
28	52.560	1.22.00	88.19	57.86	27.39	22.36	50.00	4.00	9.88	16.27	68.14	22.24	20.16	91.48	0.00	0.00
29	54.110	1.22.00	78.30	52.19	24.50	20.00	50.00	4.00	9.88	14.59	67.82	22.24	20.16	91.48	0.00	0.00
30	48.180	1.22.00	84.87	56.51	26.71	21.80	50.00	4.00	9.88	15.87	67.58	22.24	20.16	91.48	0.00	0.00
31	44.165	1.22.00	90.80	61.02	29.01	23.68	50.00	4.00	9.88	17.22	66.94	22.24	20.16	91.48	0.00	0.00
32	2.560	1.22.00	80.07	55.07	25.97	21.20	50.00	4.00	9.88	15.44	66.31	22.24	20.16	91.48	0.00	0.00

**Detailed Calculation For Upstream Protection Works for 2mx2m Single cell Box
Culvert**

ITEM NO.	ITEM	DESCRIPTION	UNIT	NO.	L (m)	B (m)	AREA (m ²)	H (m) / Thick. (m)	Volume/Qty (m ³)	SUB HEAD
	Segment 1									
5.01	Earthwork in excavations:									
	For RCC Rigid flooring (M-30)		m ³	1.00	3.00	4.40	13.20	0.60	7.92	7.92
	For Hand woven wire crates in flexible apron for protection work in upstream		m ³	1.00	3.00	5.40	16.20	1.30	21.06	21.06
	For curtain wall in segment 1		m ³	1.00	5.40	1.70		1.40	12.85	12.85
									Sub Total	41.83
	70% in Ordinary Soil									29.28
	30% in Hard Rock									12.55
5.02	PCC M-15 below curtain wall	100 mm thick.	m ³	1.00	5.40	1.70		0.10	0.92	0.92
5.02	PCC M-15 below RCC flooring	100 mm thick.	m ³	1.00	3.00	4.40	13.20	0.10	1.32	1.32
									Sub Total	2.24
5.04	RCC rigid flooring M-30			1.00	3.00	4.40	13.20	0.20	2.64	2.64
5.04	RCC Side walls over rigid flooring M-30	0.2 m thick.	m ³	2.00	3.16	0.20		1.00	1.26	1.26
									Sub Total	3.90
5.04	Rcc Curtain wall 1		m ³	1.00	5.40		0.84		4.54	4.54
5.20	Hand woven wire crates in flexible apron for	1000 mm thick.	m ³	1.00	3.00	5.40	16.20	1.00	16.20	16.20
5.07	Reinforcement in Rcc curtain wall		t	1.00	4.54	@50kg/m3 of concrete			226.80	0.227
5.07	Reinforcement in RCC flooring		t	1.00	3.90	@50kg/m3 of concrete			195.25	0.195
	Segment-2									
5.01	Earthwork in excavations:									
	Rcc slope protection structure in segment 2		m ³	1.00	1.00	3.40		0.60	2.04	2.04
	For curtain wall in segment 2		m ³	1.00	3.40	0.40		1.50	2.04	2.04
									Sub Total	2.04
	50% in Ordinary Soil									1.02
	50% in Hard rock									1.02
5.02	Pcc below Curtain wall			1.00	3.40	0.40		0.10	0.14	0.14
5.02	Pcc below Concrete for slope protection	(per meter)	m ³	1.00	1.00	3.40		0.10	0.34	0.34
									Sub Total	0.48
5.04	Rcc Concrete for slope protection structure	(per meter)	m ³	1.00	1.00	3.40		0.20	0.68	0.68
	RCC Side walls over rigid flooring M-30	(per meter)	m ³	2.00	1.00	0.20		1.00	0.40	0.40
5.04	Rcc Curtain wall		m ³	1.00	3.40	0.3		1.50	1.53	1.53
5.04	Buffle pier		m ³	28.00	0.20	0.20		0.20	0.224	0.224
									Sub Total	2.83
5.07	Reinforcement in Rcc curtain wall		t	1.00	1.5300	@50kg/m3 of concrete			76.50	0.077
5.07	Reinforcement in RCC slope protection		t	1.00	1.30	@50kg/m3 of concrete			65.20	0.065
	Segment 3									
5.01	Earthwork in excavations:									
	For catchpit		m ³	1.00	1.95	3.80	7.41	1.30	9.63	9.63
	50% in Ordinary Soil									4.82
	50% in Hard Rock									4.82
5.02	PCC M-15 below Catchpit		m ³	1.00	1.85	3.70	6.85	0.10	0.68	0.68
5.04	Rcc M-30 Catchpit walls	A1 (per meter)	m ³	2.00	1.50	0.20		1.00	0.60	0.60
		A2 (per meter)	m ³	1.00	3.40	0.20		1.00	0.68	0.68
									Sub Total	1.28
5.04	RCC M-30 Catchpit base		m ³	1.00	1.85	3.90		0.20	1.44	1.44
5.07	Reinforcement in Rcc catchpit walls		t	1.00	1.28	@50kg/m3 of concrete			64.00	0.064
	Reinforcement in Rcc catchpit base		t	1.00	1.44	@50kg/m3 of concrete			72.15	0.072

Detailed Calculation For Downstream Protection Works for 3mx3m Single cell Box Culvert

ITEM NO.	ITEM	DESCRIPTION	UNIT	NO.	L (m)	B (m)	AREA (m ²)	H (m) / Thick. (m)	Volume/Qty (m ³)	SUB HEAD TOTAL
5.1	Earthwork in excavations for protection works									
	For RCC Rigid flooring (M-30)		m ³	1.00	5.00	5.40	27.00	0.30	8.10	8.10
	For Hand woven wire crates in flexible apron for protection work		m ³	1.00	6.00	6.40	38.40	1.00	38.40	38.40
	For front curtain wall		m ³	1.00	6.40	1.70		1.30	14.14	14.14
	For back curtain wall		m ³	0.00	4.40	0.30		1.00	0.00	0.00
	70% in soil									60.64
	30% in ordinary rock									42.45
										18.19
	Outlet/downstream Protection Work									
5.04	RCC channel/flooring M-30	0.2m thick	m ³	1.00	5.00	5.40		0.20	5.40	5.40
5.04	RCC Side wall of Rigid Flooring M-30			2.00	5.10	0.20		1.00	2.04	2.04
5.04	RCC concrete block		m ³	36.00	0.30	0.20		0.30	0.65	0.65
		Sub Total								8.09
5.04	RCC in Front Curtain wall		m ³	1.00	6.40		0.81		5.18	5.18
5.04	RCC in Back Curtain wall		m ³	1.00	4.40	0.30		1.00	1.32	1.32
		Sub Total								6.50
5.02	PCC M-15 below RCC Rigid flooring	0.1m thick	m ³	1.00	5.00	5.40		0.10	2.70	2.70
5.02	PCC M-15 below front Curtain wall		m ³	1.00	6.40	1.70		0.10	1.09	1.09
5.20	Hand woven wire crates in flexible apron for protection work	1m thick	m ³	1.00	6.00	6.40		1.00	38.40	38.40
5.07	Reinforcement in RCC flooring		Mt	1.00	8.09		@50kg/m3 of concrete		404.38	0.404
5.07	Reinforcement in curtain wall		Mt	1.00	6.50		@50kg/m3 of concrete		325.20	0.325

Summary Of Quantities for bridges of PP road

Item No. as per Quantity Estimates	Item No. as per BOQ	Description	Unit	RCC Slab at KM 27.165	RCC voided Slab at KM. 31.531	Total quantity	Item No. in Rate Analysis	Remarks
1	2	3	4	5	6	7	8	9
		BILL NO. 6: NEW BRIDGES						
6.01	6.01	Excavation for Foundation of Structure, as per drawings and Technical Specification, in						
	a	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	1075	712	1787	(3.32+3.33)/2	
	b	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum			0	3.34	
	c	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum			0	3.9	
	d	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	461	305	766	3.8 A	
6.02	6.02	PCC (M 15 Grade) in Levelling course below Foundation, Pile Cap, Approach slab etc. complete as per drawings and Technical Specification Section 1700, 2100, 2500 and 2700.	cum	55	48	103	14.10	
6.03	6.03	RCC / PCC in open foundations excluding reinforcement complete as per drawings	cum			0		
	a	M 30 Grade	cum			0	12.8 G case - I	RCC rates are taken
	b	M 35 Grade	cum	353	206	559	12.8 H case - I	RCC rates are taken
6.04	6.04	RCC / PCC in Substructure, Wing Wall, retaining wall, Box Cell Structures, Seismic restrainer etc. excluding reinforcement, in following grades, complete as per Drawings and Technical Specification Section 1700 and 2200				0		
	a	M 25	cum			0	13.5 F (p) Case-I	
	b	M 30	cum			0	13.5 G (p) Case-I	
	c	M 35	cum	382	250	632	13.5 F (h) Case-I	
6.05	6.05	RCC in Superstructure excluding reinforcement complete as per drawings and Technical Specification Section 1700 and 2300.				0		
	a	Solid Slab (M 30 Grade)	cum	80	98	177	14.1C Case I (i) q	
	b	Deck Slab for Composite Girder (M35 Grade)	cum			0	14.1D Case I (ii) q	
	c	Voided Slab (M35 Grade)	cum			0	14.1D Case I (i) q	
6.06	6.06	Steel Reinforcement Fe 500D in Foundation, Substructures Superstructure etc. complete as per drawings and Technical Specification Section 1600 .						
	6.07	Steel Girder for Steel Composite Superstructure including railing and fixing of girder with Bearing complete as per drawings and Technical Specification 1000 and 1900.	MT	104	74	178	(12.40+13.6+14.2)/3	Average of three
6.08	6.08	Bituminous (Type 2) Wearing Coat as per drawings and Technical Specification Section 2700.	MT			0	14.25(i)	
6.09	6.09	40 thk. PCC (M25) finished with 15 thk plaster '1:3' complete as per drawings and Technical Specification.	sqm	128	145	272	((5.8* .040) for grading)+5.14)	
6.10	6.10	Bearings, of following Type, as per drawings and Technical Specification Section 2000	cum	2	2	5	13.5 F (p) Case-I	
	a	Tar Paper Bearings	sqm	39	39	77	As per Market	
	b	Elastomeric Bearings	cum			0	13.14	
	c	POT cum PTEE Bearings				0		

SUMMARY FOR QUANTITY CALCULATION FOR BRIDGE AT KM - 31.84

Item No.	Description	Unit	Value
6.01	for soft rock	cum	712.04
	for hard rock with no blast	cum	305.16
6.02	PCC Levelling Course (M15)	cum	48.15
6.03	RCC/PCC in open foundation	cum	205.57
6.04	RCC/PCC in Substructure	cum	249.73
6.05	RCC in Superstructure	cum	97.83
6.06	Steel Reinforcement	MT	74.24
	HYSD bar reinforcement complete as per drawings and technical specifications:		
	Abutment =	MTon	55.4
	Pier =	MTon	0.0
	Retaining wall =	MTon	8.3
	Crash Barrier @100	MTon	0.8
	Superstructure @100 =	MTon	9.8
	Total =	MTon	74.2
6.08	Bituminous Wearing coat	sqm	144.50
6.09	Wearing coat on Footpath	sqm	2.43
6.10	Tar paper bearing	sqm	38.64
6.11	Expansion joints	m	26.00
6.13	Drainage Spouts	Nos	13.00
6.14	Approach Slab	cum	33.08
6.15	Length of hand rails	Rm	34.00
6.16	Crash Barrier	cum	8.14
6.17	Earth work in Filling	cum	0.00
6.18	Weep holes	Nos	245
6.19	Backfilling behind earth retaining structures	cum	1240
6.20	Filter media behind earth retaining structures	cum	153
6.21	Boulder grouted with cement mortar	cum	121.80
6.22	PCC M-15 in annular space	cum	59.98
6.23	Preparation of rock foundation	sqm	202.50
6.24	Lettering and No. of bridges	Nos	40.00
6.25	Coats of cement paint on crash barrier and railing	sqm	340.00

Station/ Chainage (m)	Type of Retaining Structure	Retain Ht. (m)	Retain Length (m)	Culvert Position	X-Area of GW (m ²)	Surface Area of RE Wall (m ²)	Volume of GW (m ³)	RE Wall Actually Provided GW (m ²)	Add Qty GW in place replaced RE Wall (m ²)	Deducted Qty GW at Culvert location (m ³)	Exc. Qty for GW (m ³)	Backfill Qty for GW (m ³)	Exc. Qty for REW (m ³)	Backfill Qty for REW (m ³)
9320	GW	2.5	1.37		9.25	0.00	12.66				16.92	6.32	0.00	0.00
9330	GW	3.5	10.00		12.75	0.00	127.50				178.20	72.17	0.00	0.00
9340	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00
9350	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00
9360	Para W	0.0	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
9360.631	Para W	0.0	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
9370	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9380	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9390	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9391.791	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9400	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9405.291	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9410	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9420	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9420.291	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9430	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9440	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9450	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9460	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9470	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9480	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9490	GW	1.5	10.00		4.13	0.00	41.30				63.85	25.98	0.00	0.00
9500	GW	1.5	10.00		4.13	0.00	41.30				63.85	25.98	0.00	0.00
9510	GW	3.0	10.00		9.75	0.00	97.50				139.97	58.46	0.00	0.00
9520	GW	3.0	10.00		9.75	0.00	97.50				139.97	58.46	0.00	0.00
9525.675	GW	3.0	5.08		9.75	0.00	49.48				71.03	29.67	0.00	0.00
9530	GW	2.5	4.52		9.25	0.00	45.56				60.86	22.75	0.00	0.00
9540	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00
9546.504	GW	2.0	6.50		4.63	0.00	39.08				49.78	23.00	0.00	0.00
9550	GW	2.0	3.50		4.63	0.00	16.17				26.76	12.36	0.00	0.00
9560	GW	2.0	10.00		4.63	0.00	46.25				76.54	35.36	0.00	0.00
9565.719	GW	2.0	5.79		4.63	0.00	26.77				44.31	20.47	0.00	0.00
9570	GW	2.0	4.21		4.63	0.00	19.48				32.23	14.89	0.00	0.00
9580	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9590	Para W	0.0	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
9600	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9610	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9620	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9630	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9640	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9644.256	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9644.356	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9644.357	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9650	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9660	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9670	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9680	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9690	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9700	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9710	GW	1.5	10.00	C	4.13	0.00	41.30			33.04	63.85	25.98	0.00	0.00
9720	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9722.56	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9730	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9740	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9750	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9760	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9770	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9780	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9785.48	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9790	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9800	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9800.48	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9810	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9813.96	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9820	Para W	0.0	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
9830	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9840	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9850	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9860	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9870	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9880	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9890	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9900	GW	2.0	10.00		4.63	0.00	46.25				76.54	35.36	0.00	0.00
9905.892	GW	2.0	5.89		4.63	0.00	27.25				45.10	20.84	0.00	0.00
9910	GW	3.0	4.11		9.75	0.00	40.05				57.50	24.01	0.00	0.00
9920	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00
9925.178	GW	2.5	5.18		9.25	0.00	47.90				63.99	23.92	0.00	0.00
9930	GW	2.0	4.82		4.63	0.00	22.30				36.91	17.05	0.00	0.00
9940	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9946.607	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9950	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9960	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9970	Para W	0.0	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
9980	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9990	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10000	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10010	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10020	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10029.189	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10030	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10040	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10044.183	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10050	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10050.016	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10060	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10070	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10080	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10083.212	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10083.312	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10083.313	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10090	Para W	0.0	0.00											

Quantity Calculation for Traffic Signs

SL.NO.	NAME	CODE	LOCATION OF CAUTIONARY SIGNS																			
			KM36-KM37	KM37-KM38	KM38-KM39	KM39-KM40	KM40-KM41	KM41-KM42	KM42-KM43	KM43-KM44	KM44-KM45	KM45-KM46	KM46-KM47	KM47-KM48	KM48-KM49	KM49-KM50	KM50-KM51	KM51-KM52	KM52-KM53			
	STOP	14.01								1						2						
	GIVE WAY	14.02																				
	GIVE WAY TO BUSES EXISTING THE BUS BAY	14.03																				
	NO STOPPING OR STANDING	14.28																				
	NO PARKING	14.29																				
	MAXIMUM SPEED LIMIT	14.37																				
	COMPULSORY KEEP LEFT	14.48																				
	COMPULSORY KEEP RIGHT	14.49																				
	COMPULSORY SOUND HORN	14.57	2	2				2	2	2				2	2	2		2				
	LEFT HAND CURVE	15.01						2	1													
	RIGHT HAND CURVE	15.02						2														
	RIGHT HAIRPIN BEND	15.03						2	1													
	LEFT HAIRPIN BEND	15.04																				
	RIGHT HAIRPIN BEND	15.04A																				
	RIGHT REVERSE BEND	15.05	1	1	1					1				1	1	2	2	1	1			
	LEFT REVERSE BEND	15.06	1	1	1					1				1	1	2	2	1	1			
	SERIES OF BEND	15.07	2	2	2					4		2	2	4	4	2	2	2	2			
	SIDE ROAD RIGHT	15.09			2		2	2														
	SIDE ROAD LEFT	15.10			2					1		1	2									
	Y-INTERSECTION LEFT	15.11			2					1		1				2						
	Y-INTERSECTION RIGHT	15.12																				
	Y-INTERSECTION	15.13																				
	X-ROAD	15.14																				
	ROUNDABOUT	15.15																				
	T-INTERSECTION	15.17																				
	T-INT. MAJOR ROAD	15.18																				
	STAGGERED INTERSECTION RIGHT	15.20																				
	STAGGERED INTERSECTION LEFT	15.21																				
	NARROW BRIDGE AHEAD	15.25																				
	STEEP ASCENT	15.26	1		1					1												
	STEEP DESCENT	15.27	1		1					1												
	PEDESTRIAN CROSSING	15.33			2											2						
	SCHOOL AHEAD	15.34		2	2																	
	BUILT UP AREA	15.35													1	1						
	PLAYGROUND AHEAD	15.58		1	1																	
	BARRIER	15.60																				
	FALLING ROCKS	15.65																				
	SINGLE CHEVRON (SMALL)	15.72	30	29	29		32	38	41	36	41	39	43	43	29	46	39	40		35		
	DOUBLE CHEVRON	15.74																				
	OBJECT MARKER(LEFT)	15.76	4	5	7		10	9	9	6	14	12	10	9	10	8	7	8		10		
	OBJECT MARKER(RIGHT)	15.77	4	5	7		10	9	9	6	14	12	10	9	10	8	7	8		10		
	TWO WAY HAZARD MARKER	15.78																				
	MAP TYPE ADVANCE DIRECTION SIGN	16.02																				
	REASSURANCE SIGN	16.05																				
	FLAG TYPE DIRECTION SIGN	16.04		2	4											2						
	PLACE IDENTIFICATION SIGN	16.06																				
	TRUCK LAY- BY	16.07		1	1										1	1						
	WEIGH BRIDGE AHEAD	16.09																				
	GANTORY BOARD	16.10																				
	TOILET	17.05																				
	FILLING STATION	17.06																				
	HOSPITAL	17.07																				
	POLICE STATION	17.13																				
	PICNIC SITE	17.14																				
	HOME ZONE	17.21																				
	BUSTOP	17.35																				
	STATE HIGHWAY ROUTE MARKER SIGN	20.01			2											2						
	NATIONAL HIGHWAY ROUTE MARKER SIGN	20.02																				
	TOTAL NO OF SIGNS		46	57	63	54	58	67	62	59	75	68	67	69	59	84	57	62		61		

Calculation of Quantity for Overhead Gantry

S. No.	Description	Unit	No	Length (m)	Breadth (m)	Height / Depth (m)	Unit weight (Kg/m)	Quantity (Kg)
(A) Truss								
	80mm dia NB Pipe	m	4	14.10			8.36	471.50
	40mm dia NB Pipe at Side Elevation							
	Vertical Side Pipe	m	28	1.24			3.56	123.60
	Horizontal Pipe	m	4	0.74			3.56	10.54
	Diagonal pipe at ends	m	4	1.56			3.56	22.21
	Diagonal pipe at midway	m	22	1.63			3.56	127.66
	25mm dia NB Pipe at top & bottom Elevation							
	Diagonal pipe at ends	m	4	1.22			2.41	11.76
	Diagonal pipe at midway	m	22	1.30			2.41	60.93
	Cross Pipe	m	24	0.67			2.41	38.75
(B) Vertical Support of Pipe 300 mm Dia								
		m	2	5.00			49.30	493.00
Total wt (A+B)								1367.96
S. No.	Description	Unit	No	Length (m)	Breadth (m)	Height / Depth (m)	Unit weight (Kg/cum)	Quantity
(C) Gusset Plates								
	Gusset Plate at top of Truss Frame	sqm	4	1.00	1.00	0.01	7854	376.99
	Stiffners for Gusset Plate at Top frame of Truss	Sqm	8	0.56	0.30	0.01	7854	105.56
	Frame	sqm	4	1.00	1.00	0.01	7854	376.99
	Stiffners for Gusset Plate at Base of Truss Frame	Sqm	8	0.56	0.30	0.01	7854	105.56
	M.S. Plates welded to Bolts	Sqm	16	0.10	0.10	0.01	7854	12.57
Total wt (C)								977.67
Total weight of One Overhead Gantry (A+B+C) in (kg)								2345.63

Location of Proposed Gantry Signs

S. No.	Location	No. of Gantry Signs
1	Start of Project Road	1
2	End of Project Road	1
Total		2

S. No.	Description	Unit	No. of Overhead Gantry	No	Length (m)	Breadth (m)	Height / Depth (m)	Quantity	Say
1	Gantry Mounted Overhead Signs	Sqm	2	6	2.30	1.30		35.88	40
2	Foundation Pedestal								
	Excavation	cum	2	2	2.40	2.40	2.10	48.38	50
	PCC M-15 in footing	cum	2	2	2.40	2.40	0.10	2.30	3
	M-25 RCC in Footing	cum	2	2	2.20	2.20	0.30	5.81	6
	M-25 RCC in Pedestal	cum	2	2	1.00	1.00	2.45	9.80	10
	Reinforcemnt @ 60kg/cum of concrete	tonne	2					1.87	2
3	Overhead Gantry	tonne	2					4.69	5

Quantity Calculation for Roadway Indicators

No. of Curves	Length of Curve (m)	Radius of Curve (m)	Spacing of Roadway Indicators (m)	Nos. of Roadway Indicators
0	0	<21	6	0
0	0.00	21 -33	6	0
80	5545.88	33	6	1004
47	3384.05	33-41	8	470
117	8559.12	41-61	8	1187
169	12594.51	61-101	12	1219
192	16625.63	101-301	20	1023
26	2758.69	>301	25	136
Total No. of Roadway Indicators				5039

Calculation of Quantities for Junction & Median Improvement

S No.	Chainage (Km)	Type of Junction	Island in Sq mt	Area Road in Sq mt	E/S Area (Cross Leg) in Sq mt	Length of Kerb on Island and Median (Lm)	No. of Arrows		Type of Junction
							Straight	Straight-Left/Right	
1	0/000	Major	525	4425	620	547	12	4	T-Junction
2	59/030	Major	250	2175	435	270	8	0	Y-Junction
3	62/550	Major	274	2976	500	353	8	3	X-Junction
		Minor		4250	2250				
Total			1049.00	13826.00	3805.00	1170.00	28.00	7.00	

SUMMARY OF QUANTITIES

S No.	Description	unit	No	Area (sqm)	Th. (m)	Qty.	Say
1	BC	cum	1	13826.00	0.030	414.78	415.00
2	DBM	cum	1	13826.00	0.050	691.30	691.00
3	WMM	cum	1	13826.00	0.250	3456.50	3457.00
4	GSB	cum	1	13826.00	0.210	2903.46	2903.00
5	Subgrade	cum	1	17631.00	0.500	8815.50	8816.00
6	Prime Coat	sqm	1	13826.00		13826.00	13826.00
7	Tack Coat	sqm	1	13826.00		13826.00	13826.00
8	Kerb Stone in Island & Medians	m		1170.00		1170.00	1170.00
9	Soil Filling in Medians & Island	cum	1	1049.00	0.710	744.79	745.00

GSB = $4425 \times 0.21 = 929.25$

WMM = $4425 \times 0.25 = 1106.25$

DBM = $4425 \times 0.05 = 221.25$

BC = $4425 \times 0.03 = 132.75$

PC = $4425 = 4425$

TC = $4425 = 4425$

Kerb Stone = 547 m

Soil filling in Median & Island = $525 \times 0.710 = 372.75$

INPUT RATES OF PLANT, MACHINERY, LABOUR AND MATERIALS
(A) Usage Rates of Plant and Machinery

Sl. No.	Description of Machine	Activity	Output of Machine	Output	Unit	Rate (Rs.)
P&M-001	Air Compressor	General Purpose	capacity In cfm	170/250	hour	353.10
P&M-002	Batching and Mixing Plant (a) 30 cum capacity	Concrete Mixing	cum/hour	20	hour	2,361.70
P&M-003	Batching and Mixing Plant (b) 15 - 20 cum capacity	Concrete Mixing	cum/hour	13	hour	1,969.00
P&M-004	Bitumen Pressure Distributor	Applying bitumen tack coat	sqm/hour	1750	hour	1,135.20
P&M-005	Bitumen Boiler oil fired	Bitumen Spraying	capacity In litre	1500	hour	347.60
P&M-006	Concrete Paver Finisher with 40 HP Motor	Paving of concrete surface	cum / hour	20	hour	3,418.80
P&M-007	Concrete Pump of 45 & 30 cum capacity	Pumping of concrete	cum / hour	33 / 22	hour	460.90
P&M-008	Concrete Bucket	For Pouring concrete	capacity In cum	1	hour	20.90
P&M-009	Concrete Mixer (a) 0.4/0.28 cum	Concrete Mixing	cum/hour	2.5	hour	474.10
P&M-010	Concrete Mixer (b) 1 cum	Concrete Mixing	cum/hour	7.5	hour	795.30
P&M-011	Crane (a) 80 tonnes	Lifting Purpose			hour	1,936.00
P&M-012	Crane (b) 35 tonnes	Lifting Purpose			hour	1,289.20
P&M-013	Crane (c) 3 tonnes	Lifting Purpose			hour	440.00
P&M-014	Dozer D - 80 - A 12	Spreading /Cutting / Clearing	cum/hour	300/150/250	hour	3,936.90
P&M-015	Dozer D - 50 - A 15	Spreading /Cutting / Clearing	cum/hour	200/120/150	hour	2,778.60
P&M-016	Emulsion Pressure Distributor	Applying emulsion tack coat	sqm/hour	1750	hour	885.50
P&M-017	Front End loader 1 cum bucket capacity	Soil loading / Aggregate loading	cum/hour	60 /25	hour	1,016.40
P&M-018	Generator (a) 125 KVA	Generation of electric Energy	KVA	100	hour	773.30
P&M-019	Generator (b) 63 KVA	Generation of electric Energy	KVA	50	hour	436.70
P&M-020	GSB Plant 50 cum	Producing GSB	cum/hour	40	hour	1,150.60
P&M-021	Hotmix Plant - 120 TPH capacity	DBM/BM/SDC/ Premix	cum/hour	40	hour	24,777.50
P&M-022	Hotmix Plant - 100 TPH capacity	DBM/BM/SDC/ Premix	cum/hour	30	hour	22,685.30
P&M-023	Hotmix Plant - 60 to 90 TPH capacity	DBM/BM/SDC/ Premix	cum/hour	25	hour	18,141.20
P&M-024	Hotmix Plant - 40 to 60 TPH capacity	DBM/BM/SDC/ Premix	cum/hour	17	hour	14,525.50
P&M-025	Hydraulic Chip Spreader	Surface Dressing	sqm/hour	1500	hour	2,789.60
P&M-026	Hydraulic Excavator of 1 cum bucket	Soil Ordinary/ Soil Marshy/ Soil Unsuitable	cum/hour	60 /60 /60	hour	1,575.20
P&M-027	Integrated Stone Crusher 100TPH	Crushing of Spalls	TPH	100	hour	9,171.80
P&M-028	Integrated Stone Crusher 200 HP	Crushing of Spalls	TPH	200	hour	18,378.80
P&M-029	Kerb Casting Machine	Kerb Making	Rm/hour	80	hour	382.80
P&M-030	Mastic Cooker	Mastic Wearing coat	capacity In tonne	1	hour	62.84
P&M-031	Mechanical Broom Hydraulic	Surface Cleaning	sqm/hour	1250	hour	361.90
P&M-032	Motor Grader 3.35 mtr blade	Clearing /Spreading /GSB /WBM	cum/hour	200/200/50/50	hour	2,536.60
P&M-033	Mobile slurry seal equipment	Mixing and laying slurry seal	sqm/hour	2700	hour	1,067.00
P&M-034	Paver Finisher Hydrostatic with sensor control 100 TPH	Paving of DBM/ BM/SDC/ Premix	cum/hour	40	hour	2,965.60
P&M-035	Paver Finisher Mechanical 100 TPH	Paving of WMM /Paving of DLC	cum/hour	40/30	hour	1,474.00
P&M-036	Piling Rig with Bantonite Pump	0.75 m dia to 1.2 m dia Boring attachment	Rm/hour	2 to 3	hour	5,783.80
P&M-037	Pneumatic Road Roller	Rolling of Asphalt Surface	cum/hour	25	hour	1,378.30
P&M-038	Pneumatic Sinking Plant	Pneumatic Sinking of wells	cum/hour	1.5 to 2.00	hour	4,414.30
P&M-039	Pot Hole Repair Machine	Repair of potholes	cum/hour	4	hour	960.30
P&M-040	Prestressing Jack with Pump & access	Stressing of steel wires/stands			hour	144.10
P&M-041	Ripper	Scarifying	cum/hour	60	hour	46.20
P&M-042	Rotavator	Scarifying	cum/hour	25	hour	50.60
P&M-043	Road marking machine	Road marking	sqm/hour	100	hour	104.50
P&M-044	Smooth Wheeled Roller 8 tonne	Soil /BM Compaction	cum/hour	70/25	hour	555.50
P&M-045	Tandem Road Roller	Rolling of Asphalt Surface	cum/hour	30	hour	856.90
P&M-046	Tipper - 5 cum	Transportation of soil, GSB, WMM, Hotmix etc.	Capacity In cum	5.5	km	191.40
P&M-047	Tipper - 5 cum	Transportation of soil, GSB, WMM, Hotmix etc.	Capacity In cum	5.5	tonne.km	30.25
P&M-048	Tipper - 5 cum	Transportation of soil, GSB, WMM, Hotmix etc.	Capacity In cum	5.5	hour	701.80
P&M-049	Transit Mixer 4.0/4.5 cum	Transportation of Concrete Mix to site	cum/hour	4.5	hour	1,095.60

INPUT RATES OF PLANT, MACHINERY, LABOUR AND MATERIALS

P&M-050	Transit Mixer 4.0/4.5 cum	Transportation of Concrete Mix to site	cum/hour	4.5	tonne.km	42.35
P&M-051	Transit Mixer 3.0 cum	Transportation of Concrete Mix to site	cum/hour	3	hour	99.00
P&M-052	Transit Mixer 3.0 cum	Transportation of Concrete Mix to site	cum/hour	3	tonne.km	30.80
P&M-053	Tractor	Pulling	capacity in HP	50	hour	402.60
P&M-054	Tractor with Rotavator	Rate of Tractor + Rotavator			hour	422.40
P&M-055	Tractor with Ripper	Rate of Tractor 6+ Ripper			hour	433.40
P&M-056	Truck 5.5 cum per 10 tonnes	Material Transport	capacity / cum	4.5	km	149.60
P&M-057	Truck 5.5 cum per 10 tonnes	Material Transport	capacity / cum	4.5	hour	748.00
P&M-058	Truck 5.5 cum per 10 tonnes	Material Transport	capacity / cum	4.5	tonne.km	5.04
P&M-059	Vibratory Roller 8 tonne	Earth or soil / GSB / WBM	cum/hour	100/60/60	hour	1,103.30
P&M-060	Water Tanker	Water Transport	capacity In KL	6	hour	748.00
P&M-061	Water Tanker	Water Transport	capacity In KL	6	km	24.93
P&M-062	Wet Mix Plant 60 TPH	Wet Mix	cum/hour	25	hour	1,335.40
Sl. No.	Description of Machine				Unit	Rate (Rs.)
P&M-063	Air compressor with pneumatic chisel attachment for cutting hard clay.				hour	315.32
P&M-064	Batch type cold mixing plant 100-120 TPH capacity producing an average output of 75 tonne per hour				hour	4,102.56
P&M-065	Belt conveyor system				hour	683.76
P&M-066	Boat to carry atleast 20 persons				hour	371.80
P&M-067	Cement concrete batch mix plant @ 175 cum per hour (effective output)				hour	4,102.56
P&M-068	Cement concrete batch mix plant @ 75 cum per hour				hour	3,008.78
P&M-069	Cold milling machine @ 20 cum per hour				hour	2,461.31
P&M-070	Crane 5 tonne capacity				hour	579.81
P&M-071	Crane 10 tonne capacity				hour	687.23
P&M-072	Crane 15 tonne capacity				hour	816.59
P&M-073	Crane 20 tonne capacity				hour	947.10
P&M-074	Crane 40 T capacity				hour	1,076.46
P&M-075	Crane with grab 0.75 cum capacity				hour	744.15
P&M-076	Compressor with gunting equipment along with accessories				hour	1,367.52
P&M-077	Drum mix plant for cold mixes of appropriate capacity but not less than 75 tonnes/hour.				hour	711.48
P&M-078	Epoxy Injection gun				hour	443.30
P&M-079	Generator 33 KVA				hour	585.75
P&M-080	Generator 100 KVA				hour	781.28
P&M-081	Generator 250 KVA				hour	1,758.90
P&M-082	Induction, delinduction and erection of plant and equipment including all components and accessories for pneumatic method of well sinking.				hour	4,030.29
P&M-083	Joint Cutting Machine with 2-3 blades (for rigid pavement)				hour	488.40
P&M-084	Jack for Lifting 40 tonne lifting capacity.				day	519.20
P&M-085	Piling rig including double acting pile driving hammer (Hydraulic rig)				hrs	205.59
P&M-086	Plate compactor				hour	205.59
P&M-087	Snow blower equipment 140 HP @ 600 cum per hour				hour	820.05
P&M-088	Texturing machine (for rigid pavement)				hour	1,628.00
P&M-089	Truck Tractor 30 tonne capacity				hour	952.88
P&M-090	Truck Tractor 30 tonne capacity				t.km	181.50
P&M-091	Tunnel Boring machine				hour	2,344.65
P&M-092	Vibrating Pile driving hammer complete with power unit and accessories				hour	1,145.87
P&M-093	Wet Mix Plant 100 TPH				hour	2,032.80
P&M-094	Wet Mix Plant 75 TPH				hour	1,407.12

(B) Labour

Sl. No.	Description of Labour	Unit	Rate (Rs.)
L-01	Blacksmith (IInd class)	day	357.50
L-02	Blacksmith (1st class)/ Welder/ Plumber/ Electrician	day	396.00
L-03	Blaster (Stone cutter)	day	396.00
L-04	Carpenter I Class	day	396.00
L-05	Chiseller (Head Mazdoor)	day	396.00
L-06	Driller (Jumper)	day	396.00
L-07	Driver	day	467.50
L-08	Fitter	day	396.00
L-09	Mail	day	357.50
L-10	Mason (IInd class)	day	357.50
L-11	Mason (1st class)	day	396.00
L-12	Mate / Supervisor	day	440.00
L-13	Mazdoor	day	302.50
L-14	Mazdoor/Dresser (Semi Skilled)	day	330.00
L-15	Mazdoor/Dresser/Sinker (Skilled)	day	357.50
L-16	Medical Officer	day	935.00
L-17	Operator (grouting)	day	396.00
L-18	Painter I class	day	396.00
L-19	Para medical personnel	day	715.00

INPUT RATES OF PLANT, MACHINERY, LABOUR AND MATERIALS

Lead Table						
Sl. No.	Description	From	To	Lead		
				Surface Road	Katcha Track	Total
1	Earthwork	Borrow Area	Site	0.00	0.00	0.00
2	Finished Materials	HMP Plant / Batching Plant / Crushing Plant	Site	10.00	0.00	10.00
(C) Materials						
Sl. No.	Description	Unit	Rate	Carriage (Rs)	Rate (Rs.)	
M-001	Stone Boulder of size 150 mm and below at Crusher Plant	cum	2,079.00	0.00	2,286.90	
M-002	Supply of quarried stone 150 - 200 mm size for Hand Broken at site	cum	1,975.05	0.00	2,172.56	
M-003	Boulder with minimum size of 300 mm for Pitching at Site	cum	1,917.00	0.00	2,108.70	
M-004	Coarse sand at Mixing Plant	cum	5,327.00	0.00	5,859.70	
M-005	Coarse sand at Site	cum	5,918.42	0.00	6,510.26	
M-006	Fine sand at Site	cum	5,085.17	0.00	5,593.69	
M-007	Moorum at Site	cum	1,639.36	0.00	1,803.30	
M-008	Gravel/Quarry spall at Site	cum	2,782.50	0.00	3,060.75	
M-009	Granular Material or hard murrum for GSB works at site	cum	1,803.29	0.00	1,983.62	
M-010	Granular Material or hard murrum for GSB works at Mixing Plant	cum	1,803.29	0.00	1,983.62	
M-011	Fly ash conforming to IS: 3812 (Part II & I) at HMP Plant / Batching Plant / Crushing Plant	cum	6,190.45	0.00	6,809.50	
M-012	Filter media/Filter Material as per Table 300-3 (MoRT&H Specification)	cum	1,741.00	0.00	1,915.10	
Sl. No.	Description	Unit	Rate at Plant (HMP/Batching) (Rs.)		Rate at Site (Rs.)	
M-013	Close graded Granular sub-base Material 53 mm to 9.5 mm	cum	1655.00		1,820.50	
M-014	Close graded Granular sub-base Material 37.5 mm to 9.5 mm	cum	1675.00		1,842.50	
M-015	Close graded Granular sub-base Material 26.5 mm to 9.5 mm	cum	1711.00		1,882.10	
M-016	Close graded Granular sub-base Material 9.5 mm to 4.75 mm	cum	1736.00		1,909.60	
M-017	Close graded Granular sub-base Material 9.5 mm to 2.36 mm	cum	1741.00		1,915.10	
M-018	Close graded Granular sub-base Material 4.75mm to 2.36 mm	cum	1746.00		1,920.60	
M-019	Close graded Granular sub-base Material 4.75mm to 75 micron mm	cum	1749.00		1,923.90	
M-020	Close graded Granular sub-base Material 2.36 mm	cum	1751.00		1,926.10	
M-021	Stone crusher dust finer than 3mm with not more than 10% passing 0.075 sieve	cum	1639.36		1,803.30	
M-022	Coarse graded Granular sub-base Material 2.36 mm & below	cum	2561.27		2,817.40	
M-023	Coarse graded Granular sub-base Material 4.75mm to 75 micron mm	cum	2556.27		2,811.90	
M-024	Coarse graded Granular sub-base Material 4.75 mm to 2.36 mm	cum	2551.27		2,806.40	
M-025	Coarse graded Granular sub-base Material 9.5 mm to 4.75 mm	cum	2536.27		2,789.90	
M-026	Coarse graded Granular sub-base Material 26.5 mm to 4.75 mm	cum	2511.27		2,762.40	
M-027	Coarse graded Granular sub-base Material 26.5 mm to 9.5 mm	cum	2506.27		2,756.90	
M-028	Coarse graded Granular sub-base Material 37.5 mm to 9.5 mm	cum	2481.27		2,729.40	
M-029	Coarse graded Granular sub-base Material 53 mm to 26.5mm	cum	2431.27		2,674.40	
M-030	Aggregates below 5.6 mm	cum	2025.00		2,227.50	
M-031	Aggregates 22.4 mm to 2.36 mm	cum	1819.00		2,000.90	
M-032	Aggregates 22.4 mm to 5.6 mm	cum	1816.00		1,997.60	
M-033	Aggregates 45 mm to 2.8 mm	cum	1763.00		1,939.30	
M-034	Aggregates 45 mm to 22.4 mm	cum	1741.00		1,915.10	
M-035	Aggregates 53 mm to 2.8 mm	cum	1773.00		1,950.30	
M-036	Aggregates 53 mm to 22.4 mm	cum	1761.00		1,937.10	
M-037	Aggregates 63 mm to 2.8 mm	cum	1758.00		1,933.80	
M-038	Aggregates 63 mm to 45 mm	cum	1589.00		1,747.90	
M-039	Aggregates 90 mm to 45 mm	cum	1559.00		1,714.90	
M-040	Aggregates 10 mm to 5 mm	cum	1999.00		2,198.90	
M-041	Aggregates 11.2 mm to 0.09 mm	cum	1997.00		2,196.70	
M-042	Aggregates 13.2 mm to 0.09 mm	cum	1995.00		2,194.50	
M-043	Aggregates 13.2 mm to 5.6 mm	cum	1990.00		2,189.00	
M-044	Aggregates 13.2 mm to 10 mm	cum	1970.00		2,167.00	
M-045	Aggregates 20 mm to 10 mm	cum	1817.00		1,998.70	
M-046	Aggregates 25 mm to 10 mm	cum	1812.00		1,993.20	
M-047	Aggregates 19 mm to 6 mm	cum	1829.00		2,011.90	
M-048	Aggregates 37.5 mm to 19 mm	cum	1782.00		1,960.20	
M-049	Aggregates 37.5 mm to 25 mm	cum	1777.00		1,954.70	
M-050	Aggregates 6 mm nominal size	cum	2010.00		2,211.00	
M-051	Aggregates 10 mm nominal size	cum	1980.00		2,178.00	
M-052	Aggregates 13.2/12.5 mm nominal size	cum	1970.00		2,167.00	
M-053	Aggregates 20 mm nominal size	cum	1817.00		1,998.70	
M-054	Aggregates 25 mm nominal size	cum	1807.00		1,987.70	
M-055	Aggregates 40 mm nominal size	cum	1741.00		1,915.10	

INPUT RATES OF PLANT, MACHINERY, LABOUR AND MATERIALS

Sl. No.	Description	Unit	Rate (Rs.)
M-056	AC pipe 100 mm dia	metre	381.15
M-057	Acrylic polymer bonding coat	litre	473.55
M-058	Aluminium Paint	litre	343.20
M-059	Aluminium alloy plate 2mm Thick	sqm	984.50
M-060	Aluminium alloy/galvanised steel	tonne	1,60,854.38
M-061	Aluminium sheeting fixed with encapsulated lens type reflective sheeting Including 2% towards lettering, cost of angle iron, cost of drilling holes, nuts, bolts etc.and signs as applicable	sqm	10,781.10
M-062	Aluminium studs 100 x 100 mm fitted with lense reflectors	nos	1,113.20
M-063	Barbed wire	kg	457.38
M-064	Bearing (Cost of parts)	nos	94,915.43
M-065	Bearing (Cast steel rocker bearing assembly of 250 tonne)	nos	1,00,273.31
M-066	Bearing (Elastomeric bearing assembly consisting of 7 internal layers of elastomer bonded to 6 nos. Internal reinforcing steel laminates by the process of vulcanisation,)	nos	95,711.00
M-067	Bearing (Forged steel roller bearing of 250 tonne	nos	93,984.33
M-068	Bearing (Pot type bearing assembly consisting of a metal piston supported by a disc, PTFE pads providing sliding surfaces against stainless steel mating together with cast steel assemblies/ fabricated structural steel assemblies duly painted with all components	nos	30,800.00
M-069	Bearing (PTFE sliding plate bearing assembly of 80 tonnes)	nos	10,521.50
M-070	Bearing (Supply of sliding plate bearing of 80 tonne)	nos	10,521.50
M-071	Bentonite	kg	197.51
M-072	Binding wire	kg	79.75
M-073	Bitumen (Cationic Emulsion)	tonne	81,331.48
M-074	Bitumen (50-70 grade)	tonne	35,934
M-077	Bitumen (emulsion)	tonne	31,428
M-078	Bitumen (modified graded)	tonne	39,455
M-079	Brick	each	15.69
M-080	C.I.shoes for the pile	kg	62.37
M-081	Cement	tonne	8,629.20
M-082	Thermo Mechanically Treated bars (HYSD/TMT Bars)	tonne	59,859.20
M-083	Collar for joints 300 mm dia	nos	959.81
M-084	Compressible Fibre Board(20mm thick)	sqm	1,029.11
M-085	Connectors/ Staples	each	65.84
M-086	Copper Plate(12m long x 250mmwide)	kg	43,340.00
M-087	Corrosion resistant Structural steel	tonne	93,500.00
M-088	Corrugated sheet, 3 mm thick, "Thrie" beam section railing	kg	86.35
M-089	Credit for excavated rock found suitable for use	cum	127.05
M-090	Curing compound	litre	65.84
M-091	Delineators from ISI certified firm as per the standard drawing given in IRC - 79	each	1,023.00
M-092	Earth Cost or compensation for earth taken from private land	cum	66.55
M-093	Elastomeric slab seal expansion joint assembly manufactured by using chloroprene, elastomer for elastomeric slab unit conforming to clause 915.1 of IRC: 83 (part II),	metre	13,543.53
M-094	Electric Detonators @ 1 detonator for 1/2 gelatin stick of 125 gms each	100 nos	330.00
M-095	Epoxy compound with accessories for preparing epoxy mortar	kg	609.95
M-096	Epoxy mortar	kg	499.07
M-097	Epoxy primer	kg	438.35
M-098	Epoxy resin-hardner mix for prime coat	kg	367.07
M-099	Flag of red color cloth 600 x 600 mm	each	72.60
M-100	Flowering Plants	each	18.70
M-101	Galvanised MS flat clamp	nos	20.79
M-102	Galvanised steel wire crates of mesh size 100 mm x 100 mm woven with 4mm dia. GI wire in rolls of required size.	sqm	401.23
M-103	Galvanised structural steel plate 200 mm wide, 6 mm thick, 24 m long	kg	82.45
M-104	Gelatin 80%	kg	335.39
M-105	Geo grids	sqm	29.04
M-106	Geomembrane	sqm	35.09
M-107	Geonets	sqm	41.14
M-108	Geotextile	sqm	27.50
M-109	Geotextile filter fabric	sqm	90.09
M-110	GI bolt 10 mm Dia	nos	20.79
M-111	Grouting pump with agitator	hour	387.20
M-112	Grass (Doob)	kg	77.00
M-113	Grass (Fine)	kg	96.80
M-114	HDPE pipes 75mm dia	metre	422.40
M-115	HDPE pipes 90mm dia	metre	580.80
M-116	Hedge plants	each	12.10
M-117	Helical pipes 600mm diameter	metre	1,522.40
M-118	Hot applied thermoplastic compound	litre	860.75
M-119	HTS strand	tonne	85,391.11
M-120	Joint Sealant Compound	kg	612.11
M-121	Jute netting, open weave, 2.5 cm square opening for seedling and Mulching	sqm	660.00
M-122	LDO for steam curing	litre	126.83
M-123	M.S. Clamps	nos	16.94
M-124	M.S. Clamps	kg	85.25
M-125	M.S.shoes @ 35 Kg per pile of 15 m	kg	78.38
M-126	Mild Steel bars	tonne	58,410.00
M-127	Modular strip/box seal expansion joint including anchorage catering to a horizontal movement beyond 70 mm and upto 140mm assembly comprising of edge beams, central beam,2 modules chloroprene seal, anchorage elements, support and control system, all steel sections protected against corrosion and installed by the manufacturer or his authorised representative	metre	33,550.00

CHAPTER-1 CARRIAGE OF MATERIALS						
Sr No	Description	Unit	Quantity	Rate Rs	Cost Rs	Remarks/ Input ref.
1.1	Loading and Unloading of Stone Boulder/Stone aggregates/Sand/Kanker/Moorum.	cum				
	Placing tipper at loading point, loading with front end loader, dumping, turning for return trip, excluding time for haulage and return trip					
	Unit = cum					
	Taking output = 5.5 cum					
	Time required for					
	i) Positioning of tipper at loading point		1 Min			
	ii) Loading by front end loader 1 cum bucket capacity @ 25 cum per hour		13 Min			
	iii) Maneuvering, reversing, dumping and turning for return		2 Min			
	iv) Waiting time, unforeseen contingencies etc		4 Min			
	Total		20 Min			
	a) Machinery					
	Tipper 5.5 tonnes capacity	hour	0.33	701.80	231.59	P&M-048
	Front end-loader 1 cum bucket capacity @ 25 cum/hour	hour	0.33	1016.40	335.41	P&M-017
	b) Overhead charges @ 8 % on (a)				45.36	
	c) Contractor's profit @ 15 % on (a+b)				91.85	
	Cost for 5.5 cum = a+b+c				704.22	
	Rate per cum = (a+b+c)/ 5.5				128.04	
	Note Unloading will be by tipping.			say	128.00	
1.2	Loading and Unloading of Boulders by Manual Means					
	Unit = cum					
	Taking output = 5.5 cum					
	a) Labour					
	Mate	day	0.11	440.00	48.40	L-12
	Mazdoor for loading and unloading	day	0.75	302.50	226.88	L-13
	b) Machinery	day	0.75	701.80	526.35	P&M-048
	c) Overhead charges @ 8 % on (a+b)				64.13	
	d) Contractor's profit @ 15 % on (a+b+c)				129.86	
	Cost for 5.5 cum = a+b+c+d				995.62	
	Rate per cum = (a+b+c+d)/5.5				181.02	
	Note Unloading will be by tipping.			say	181.00	
1.3	Loading and Unloading of Cement or Steel by Manual Means and Stacking.					
	Unit = tonne					
	Taking output = 10 tonnes					
	a) Labour					
	Mate	day	0.08	440.00	35.20	L-12
	Mazdoor for loading and unloading	day	2.00	302.50	605.00	L-13
	b) Machinery					
	Truck 10 tonne capacity	hour	2.00	748.00	1496.00	P&M-057
	c) Overhead charges @ 8 % on (a+b)				170.90	
	d) Contractor's profit @ 15 % on (a+b+c)				346.06	
	Cost for 10 tonnes = a+b+c+d				2653.16	
	Rate per tonnes = (a+b+c+d)/10				265.32	
				say	265.00	
1.4	Cost of Haulage Excluding Loading and Unloading					
	Haulage of materials by tipper excluding cost of loading, unloading and stacking.					
	Unit = t.km					
	Taking output 10 tonnes load and lead 10 km = 100 t.km					
(I)	Surfaced Road					
	Speed with load : 25 km / hour.					
	Speed while Returning empty : 35 km / hour.					
	a) Machinery.					
	Tipper 10 tonne capacity					
	Time taken for onward haulage with load	hour	0.40	701.80	280.72	P&M-048
	Time taken for empty return trip.	hour	0.29	701.80	203.52	P&M-048
	b) Overhead charges @ 8 % on (a)				38.74	
	c) Contractor's profit @ 15 % on (a+b)				78.45	
	cost for 100 t km = a+b+c				601.43	
	Rate per t.km = (a+b+c)/100				6.01	
				say	6.00	
1.4	(II) Unsurfaced Graveled Road					
	Speed with load: 20 km / hour					
	Speed for empty return trip : 30 km / hour					
	a) Machinery					

CHAPTER-2 SITE CLEARANCE							
Sr No		Description	Unit	Quantity	Rate Rs	Cost Rs	Remarks/ Input ref.
2.1		Cutting of Trees, including cutting of Trunks, Branches and Removal					
		Cutting of trees, including cutting of trunks, branches and removal of stumps, roots, stacking of serviceable material with all lifts and up to a lead of 1000 metres and earth filling in the depression/nit					
		Unit = Each					
	(i)	Girth from 300 mm to 600 mm					
		a) Labour					
		Mate	day	0.02	440.00	8.80	L-12
		Mazdoors for cutting trees including cutting, refilling, compaction of backfilling and stacking of serviceable materials within 1000 metres lead by manual means.	day	0.60	302.50	181.50	L-13
		b) Machinery					
		Tractor-trolley	hour	0.10	402.60	40.26	P&M-053
		c) Overhead charges @ 8 % on (a+b)				18.44	
		d) Contractor's profit @ 15 % on (a+b+c)				37.35	
		Rate for each tree = a+b+c+d				286.36	
					say	286.00	
2.1	(ii)	Girth from 600 mm to 900 mm					
		a) Labour					
		Mate	day	0.04	440.00	17.60	L-12
		Mazdoors for cutting trees including cutting, refilling, compaction of backfilling, and stacking of serviceable materials within 1000 metres lead by manual means	day	0.90	302.50	272.25	L-13
		b) Machinery					
		Tractor-trolley	hour	0.30	402.60	120.78	P&M-053
		c) Overhead charges @ 8 % on (a+b)				32.85	
		d) Contractor's profit @ 15 % on (a+b+c)				66.52	
		Rate for each tree = a+b+c+d				510.00	
					say	510.00	
2.1	(iii)	Girth from 900 mm to 1800 mm					
		a) Labour					
		Mate	day	0.08	440.00	35.20	L-12
		Mazdoors for cutting trees including cutting, refilling, compaction of backfilling and stacking of serviceable materials within 1000 metres	day	2.00	302.50	605.00	L-13
		b) Machinery					
		Tractor-trolley	hour	0.40	402.60	161.04	P&M-053
		c) Overhead charges @ 8 % on (a+b)				64.10	
		d) Contractor's profit @ 15 % on (a+b+c)				129.80	
		Rate for each tree = a+b+c+d				995.14	
					say	995.00	
2.1	(iv)	Girth from 1800 mm to 2700mm					
		a) Labour					
		Mate	day	0.16	440.00	70.40	L-12
		Mazdoors for cutting trees including cutting, refilling, compaction of backfilling and stacking of serviceable materials within 1000 metres	day	4.00	302.50	1210.00	L-13
		b) Machinery					
		Tractor-trolley	hour	0.60	402.60	241.56	P&M-053
		c) Overhead charges @ 8 % on (a+b)				121.76	
		d) Contractor's profit @ 15 % on (a+b+c)				246.56	
		Rate for each tree = a+b+c+d				1890.27	
					say	1890.00	